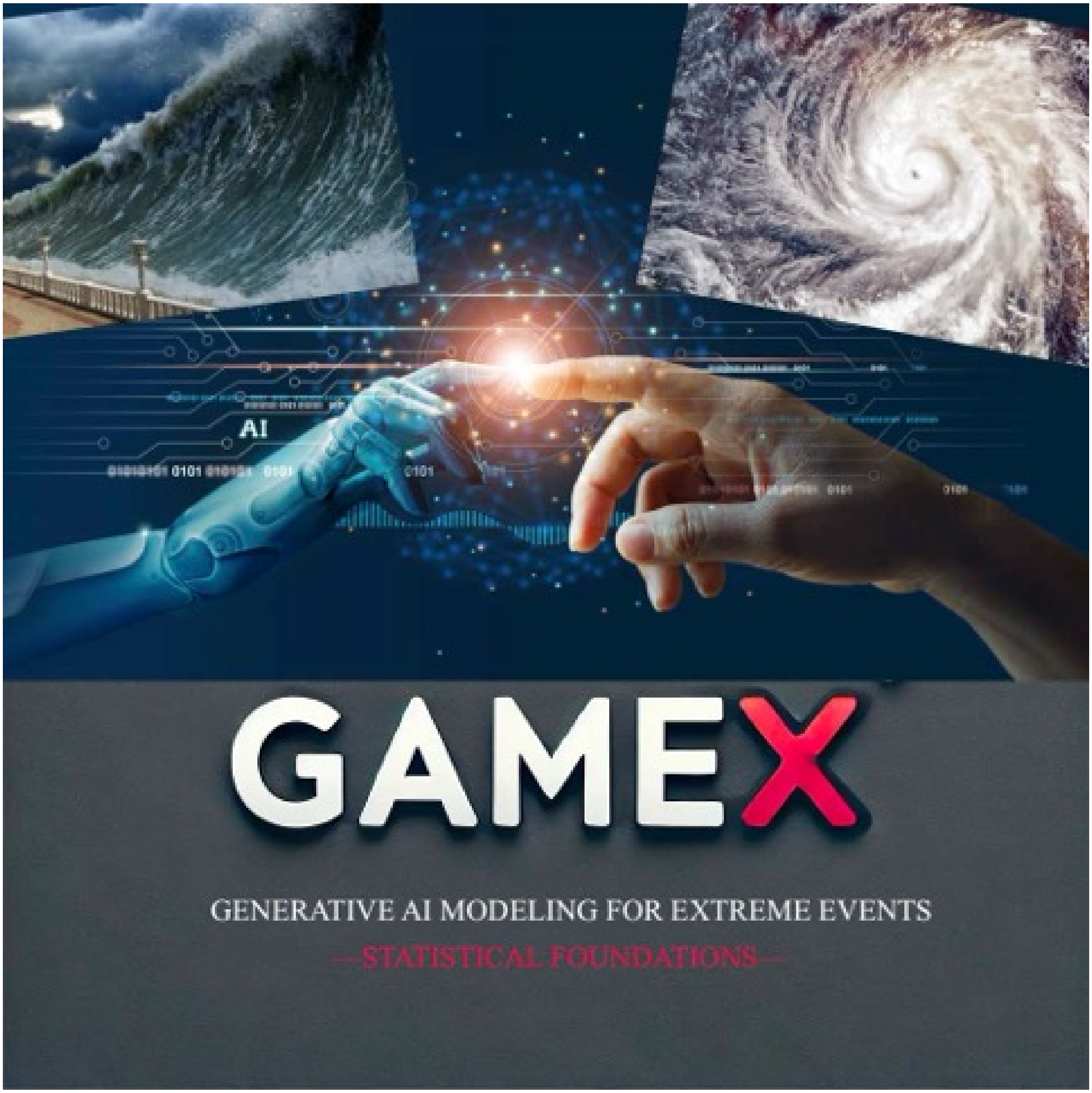
# Edinburgh Summer School on Generative AI for Extremes




**Provisional Programme—updated 1 September 2026**
**8–11 September 2026**

`https://gamex-network.github.io/school/`

**GL(EN)$^2$ (Glasgow–Lancaster–Edinburgh–Newcastle) Extremes Network**


**GAMEX Project Members:** Luca Trapin • Miguel de Carvalho • Massimo Ventrucci • Ivo Bonfanti • Johnny Lee • Giorgia Ramponi
Till Freihaut • Brendan Murphy • Luis Gutierrez Inostroza • Clemente Ferrer

## Chair

**Professor Miguel de Carvalho**
Chair of Statistics & Data Science
Elected Fellow of Generative AI Laboratory

## Organisers

Johnny Lee (Edinburgh), Lambert De Monte (Edinburgh, KAUST), Jordan Richards (Edinburgh), Daniela Castro-Camilo (Glasgow), Mengran Li (Glasgow), Teng Wei Yeo (Glasgow), and Luca Trapin (Bologna).

## Welcome

Building on the success of GAME 2025 (`https://edin.ac/4akncn7`) in Edinburgh, the first worldwide workshop on generative AI modelling for extreme events, this summer school is led by GL(EN)$^2$ (Glasgow–Lancaster–Edinburgh–Newcastle) Extremes Network and brings PhD students and early-career researchers to the interface of generative AI and extreme value theory. It is an official ECAS course (European Courses in Advanced Statistics).

## Partners

## Further Reading

In addition to the materials made available by the instructors, we recommend selected parts of the *Handbook of Statistics of Extremes* (`https://extremestats.github.io/`) and parts of the recent special issue on *Bridging Heavy Tails and AI* (`https://edin.ac/4akncn7`).

## Day 1: Tuesday — 8 September

**Theme:** *Foundations and Preparations for Generative Modelling*
**Moderator:** Luca Trapin

| Time | Session | Lead |
|---|---|---|
| 09:00–09:25 | Registration & Welcome Coffee | Organisers |
| 09:25–09:30 | *Welcome & Overview* | Chair |
| 09:30–11:00 | *Introducing Extreme Value Theory* | Miguel de Carvalho |
| 11:00–11:30 | Coffee break | |
| 11:30–12:30 | *On Neural Generative Models* | Miguel de Carvalho |
| 12:30–14:00 | Lunch | |
| 14:00–15:00 | *Probabilistic Aspects of Diffusions* | Sotirios Sabanis |
| 15:00–15:30 | Coffee break | |
| 15:30–17:00 | *Practical: Neural Generative Models in Practice* | Johnny Lee |

## Day 2: Wednesday — 9 September

**Theme:** *Generative Modelling for Extremes and Risk*
**Moderator:** TBA

| Time | Session | Lead |
|---|---|---|
| 09:30–10:30 | *GAN-based Generative Approaches for Extremes* | Stéphane Girard (online) |
| 10:30–11:00 | Coffee break | |
| 11:00–12:00 | *GAN-based Modelling for Operational Risk* | Luca Trapin |
| 12:00–13:30 | Lunch | |
| 13:30–15:30 | *Practical: Generative Modelling for Extremes and Risk* | Jordan Richards |
| 15:30–16:00 | Coffee break | |
| 16:00–16:10 | Group photo | |
| 16:10–17:10 | *Open Practice & Office Hours* | Instructors |

## Day 3: Thursday — 10 September

**Theme:** *Multivariate Extremes, Cascades, and Simulation*
**Moderator:** Giorgia Ramponi

| Time | Session | Lead |
|---|---|---|
| 09:30–10:30 | *Cascades, Multivariate Extremes, and Transformers* | Miguel de Carvalho |
| 10:30–11:00 | Coffee break | |
| 11:00–12:00 | *Multivariate Angular Simulation* | Emma Simpson |
| 12:00–13:30 | Lunch | |
| 13:30–15:30 | *Practical: Multivariate and Angular Simulation* | Clemente Ferrer |
| 15:30–16:00 | Coffee break | |
| 16:00–17:00 | *Open Practice & Office Hours* | Instructors |

## Day 4: Friday — 11 September

**Theme:** *Other Frontier Topics*
**Moderator:** Emma Simpson

| Time | Session | Lead |
|---|---|---|
| 09:30–10:30 | *Reinforcement Learning* | Giorgia Ramponi |
| 10:30–11:00 | Coffee break | |
| 11:00–12:00 | *Pre-trained Extreme Value Modelling* | Jordan Richards |
| 12:00–12:10 | *Closing Remarks* | Chair |
| 12:10–13:40 | Lunch | |

† Lectures: Lecture Theatre B; practical sessions: Teaching Studio 3217.